DRAFT

Waves

Ganesh Sundaram, Dept. of Physics, Amrita Vishwa Vidyapeetham

Contents

0.1 Course Outcomes and Syllabus 2

1 Simple Harmonic Motion **5**

1.1 Introduction 5

1.1.1 Basic Terminology 6

1.1.2 Relation with the uniform circular motion or uniformly rotating vector, rotating vector representation 7

1.2 Kinematics of SHM 8

1.3 The Spring-Mass System, its ODE and Solutions 11

1.3.1 Solution to the ODE of Spring-Mass System – Method 1 12

1.4 Kinetic, Potential and Total Energies and the Energy Method 16

1.4.1 Kinetic, potential and total energies, average kinetic and potential energies 16

1.4.2 The energy method of solution to equation of motion. 16

1.5 Simple harmonic motions and natural frequencies of simple physical systems 17

1.5.1 Simple pendulum 17

1.5.2 Elastic bar 17

1.5.3 Oscillations of piston in a cylinder filled with air 18

1.5.4 LC circuit 20

1.6 Harmonic approximation of general potentials near minima 20

1.7 Damped Oscillations, Quality Factor 22

1.8 Driven Oscillations and Resonance 22

1.9 Linearity and Superposition Principle 22

DRAFT

DRAFT

Ganesh Sundaram, Dept. of Physics, Amrita Vishwa Vidyapeetham

Waves

0.1 Course Outcomes and Syllabus

Description

Building upon the first introductory course on mechanics, this course is the first part of the two courses that introduces advanced techniques in mechanics covering topics of vector kinematics and dynamics, energy methods, momentum and angular momentum, central force and damped and forces oscillatory motion, coupled oscillations and mechanical waves in elastic media from an intermediate level book on mechanics.

Course Outcomes

On successful completion of this course, students shall be able to

1. Apply the principles of simple harmonic motion to simple physical systems,
2. Apply principles of mechanics to derive equations of coupled oscillations and mechanical waves.
3. Apply principles of mechanics to arrive at wave equations in a string and deduce its be- haviour and associated properties
4. Apply principles of ray optics to calculate images formation in mirror and lens systems and other related parameters of optical systems.
5. Apply the concepts in interference and diffraction to calculate interference fringe pattern parameters in Newton's rings and other optical systems, calculate diffraction intensity pat- terns, the resolving power of a few optical systems, apply the concepts of polarization to intensities of light passing through polarizers, optical rotation and other related parameters.

DRAFT

DRAFT

Syllabus

Unit 1

Small oscillations in bound systems: Harmonic approximation, linearity, superposition principle; damped, over damped and critically damped oscillations, relaxation time, use of complex variables; steady-state response of a damped spring-mass system under sinusoidal forces, phase lag, resonance, Q-value, absorption bandwidth, amplification, impedance in mechanical and electrical systems. [8+2] (Ref.1, 5)

Unit 2

Transverse oscillations: small oscillation approximations; Free oscillations of systems with two degrees of freedom – 2D harmonic oscillator, longitudinal or transverse oscillations two coupled masses or coupled pendulums, normal modes. Linear chain of coupled pendulums: semi-Propagation of undamped free waves, dispersion relation, mechanical band-pass filter. (Ref.4, 5) [6+2]

Unit 3

Free oscillations in systems with many degrees of freedom: Transverse waves in strings, wave equation, boundary conditions, standing waves, harmonics, wave velocity, wave number, and dispersion relation; general motion and Fourier series. (Ref.4) [3+1]

Travelling solutions, superposition of waves, beats, phase and group velocities, bandwidth theorem (without proof). Forced oscillations and characteristic impedance of a travelling wave, boundary conditions for reflection and transmission; reflection and transmission coefficients (results only); energy flux, reflection and transmission of energy – intensity coefficients (results only); impedance matching and its importance in electrical circuits. (Ref.5) [6+2]

Unit 4

Geometrical Optics: Fermat's principle and Snell's laws, total internal reflection, prisms, general theory of image formation, plane and spherical mirrors, thick and thin lenses and lens combinations, aberrations, cardinal points, optical systems. [8+2]

DRAFT

DRAFT

Unit 5

Interference: coherence; interference by division of wave front: Young's double slit, Fresnel's bi-prism; interference by division of amplitude: thin-films, Newton's rings, basic ideas of Michelson and Fabry-Perot interferometers. [4+1]

Fraunhofer diffraction; Rayleigh criterion and resolving power; diffraction gratings. [3+1]

Polarization: linear and circular polarization; double refraction and optical rotation, polarizers and quarter waveplate, Malus' law. [1.5+0.5]

References

1. Vibrations and Waves, A.P. French
2. Berkeley Physics Vol. 3, Waves. Wiley India.
3. H.J. Pain, Vibrations and Waves, 6th Edition.
4. EM waves & Optics: Ch. 34-37 of Halliday, Resnick, & Walker, Fundamentals of Physics, 10th Ed. Or equivalent material in Eugene Hecht.
5. Ajoy Ghatak, Optics, 7th Ed., TMH

Evaluation Policy

Evaluation Pattern			Weight
Continuous Evaluation	Assignments	20%	30%
	A number of quizzes/tests	10%	
Mid-term			30%
End-Semester			40%

DRAFT

DRAFT

Ganesh Sundaram, Dept. of Physics, Amrita Vishwa Vidyapeetham

Chapter 1

Simple Harmonic Motion

Contents

1.1 Introduction 5
1.2 Kinematics of SHM 8
1.3 The Spring-Mass System, its ODE and Solutions 11
1.4 Kinetic, Potential and Total Energies and the Energy Method 16
1.5 Simple harmonic motions and natural frequencies of simple physical systems 17
1.6 Harmonic approximation of general potentials near minima 20
1.7 Damped Oscillations, Quality Factor 22
1.8 Driven Oscillations and Resonance 22
1.9 Linearity and Superposition Principle 22

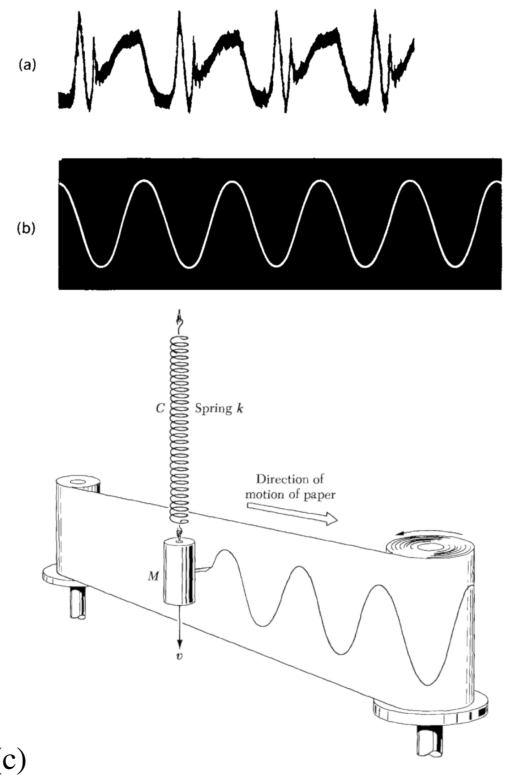
1.1 Introduction

- Oscillations and waves are among the most important subjects in physics.
  - ◇ Many systems we see in everyday life undergo periodic, repetitive motion in time as well as space.
- Key functions of life and our interactions with external world involve a nearly periodic or a cyclic motion.
  - ◇ Key essential functions: Hearts beat, lungs oscillate, legs oscillate as we move.
  - ◇ Hearing and speaking: Eardrums and larynges vibrate; vibrating air molecules travel as sound waves, create vibrations in eardrums.
  - ◇ Seeing: light waves which enables us to see is a propagating oscillations of electric and magnetic fields.
- Oscillation: It is a bounded periodic repetitive to and fro motion about a fixed equilibrium or stationary position.
  - ◇ A periodic motion, described by the variable$x(t)$, is said to have a period T if $x(t)$ repeats after some fixed time interval T:

$$x(t+T) = x(t).$$

- A special case of a periodic motion is the simple harmonic motion. It has the form

$$x(t) = A\sin\omega t \qquad \text{or} \qquad A\cos\omega t.$$



(c)

Fig. 1.1: Periodic oscillations. (a) Oscillations in heart muscles. (b) Oscillation of a tuning fork. (c) A pen attached to an oscillating spring–mass system traces its simple harmonic motion on a moving strip of paper.

DRAFT

DRAFT

Examples:

- ◇ mass oscillating in a spring-mass system
- ◇ pendulum bob oscillating about the vertical position,
- ◇ current in an LC-circuit

- Wave:
  - ◇ A wave is an oscillation traveling through space.
  - ◇ Neighboring points oscillate with their phases having certain relationship.
  - ◇ Examples: waves in a string; sound waves propagating though air, solids or liquids; light (electromagnetic oscillations) propagating through vacuum.

1.1.1 Basic Terminology

- **Period** and the periodicity of motion. As mentioned earlier, in a periodic motion, the variable $x(t)$ is said to oscillate with period T if $x(t)$ repeats itself after some fixed time interval T:

$$x(t+T) = x(t). \tag{1.1}$$

(Said in words, in mathematical terms, x at time $t+T$ is the same as x at time t for all t).

- ◇ This means that in T seconds (or some unit of time), the motion completes one cycle or period of the motion.

- Let's recall a form of simple harmonic motion,

$$x(t) = A\sin\omega t. \tag{1.2}$$

- Question that arises naturally is that how this periodic and how is the constant parameter ω is related to T.
- Applying the definition of periodicity in Eq.(1.1) to the SHM in Eq.(1.2), we get

$$A\sin\omega(t+T) = A\sin\omega t \quad \Rightarrow \quad A\sin(\omega t+\omega T) = A\sin\omega t \quad \Rightarrow$$

$$\omega T = 0, \pm 2\pi, \pm 4\pi, \ldots$$

Our definition of Tcorresponds to the smallest positive non-zero interval for which Eq.(1.1) holds. Hence we arrive at

$$\omega T = 2\pi.$$

DRAFT

DRAFT

- **Frequency**. To proceed further, let us recall that the number of cycles (of oscillation) per unit time is simply

  $$\#\text{ cycles of oscillation per unit time} = \frac{1}{T} \equiv \nu \equiv \text{frequency of the oscillation.}$$

- **Angular frequency.** Thus, the parameter ω is

  $$\omega = \frac{2\pi}{T} = 2\pi\nu = 2\pi \times (\text{number of cycles per unit time}) \equiv \text{angular frequency.}$$

  Why does angle enter into the discussions here? We will discuss this below in detail. In brief, one cycle of motion corresponds to one cycle of uniform circular motion which is equivalent to a rotation by angle 2π radians. The argument of the sine function, ωt, increases by 2π in one cycle.

- **Phase** of oscillation is simply the argument of the sine function:

  $$\text{phase of oscillation} \equiv \omega t \equiv \theta.$$

  Its physical significance will become clear when we discuss its analogy with uniform circular motion.

- **Amplitude**. For the SHM given in Eq.(1.2), the maximum and minimum values of the variable x are A and $-A$. We say that x varies in the interval $[-A, A]$, while t can take any real values – past is represented by negative values of t. The quantity A denotes the amplitude of the oscillation.

1.1.2 Relation with the uniform circular motion or uniformly rotating vector, rotating vector representation

- *Simple harmonic motions projections of a uniform circular motion.*

  ◇ Consider a point moving on a circle of radius A, starting at $(A, 0)$ at time $t = 0$, moving with uniform speed $ds/dt = v$. If θ is the angle of the vector $\vec{r} = x\hat{i} + y\hat{j}$ with x-axis then

  $$x = A\cos\theta, \qquad y = A\sin\theta.$$

  ◇ The rate of change of angle is then, by $R\mathrm{d}\theta = \mathrm{d}s$,

  $$\frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{\mathrm{d}s/\mathrm{d}t}{R} = \frac{v}{A} \equiv \omega \qquad \text{(angular frequency).}$$

  ◇ With $\theta(0) = 0$, we get $\theta = \omega t$. Thus, position vector of a uni-

DRAFT

DRAFT

formly rotating vector is

$$\vec{r}(t) = A\cos\omega t\,\hat{i} + A\sin\omega t\,\hat{j} = A\left[\cos\theta(t)\,\hat{i} + \sin\theta(t)\,\hat{j}\right].$$

The arguments $\omega t = \theta$ represents the angle of the vector with x-axis.

- In one circular motion, the vector rotates by angle 2π radians.
- Thus, variations represented by $A\cos\omega t$ and $A\sin\omega t$ represents vertical (x) and horizontal (y) projections of a uniform circular motion.
- A more general case is shown in the figure, with $\theta = \omega t + \alpha$.

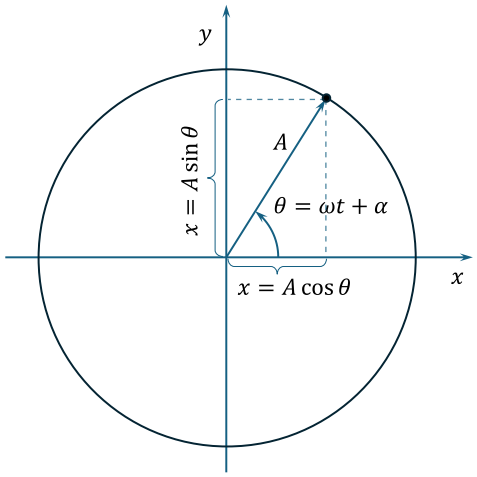


Fig. 1.2: Two simple harmonic motions from projections of a uniform circular motion. This is known as the rotating vector $\vec{r} = x\hat{i} + y\hat{j}$ or phasor representation.

- **Phasor or a rotating complex vector representation**.
  - Calculations become amazingly easier if we work with complex numbers instead.
  - Let us define a complex number with x- and y- projections of a circular motion as real and imaginary parts:

$$z = x + iy = A\cos\omega t + iA\sin\omega t = A(\cos\omega t + i\sin\omega t) = Ae^{i\omega t} = Ae^{i\theta(t)}.$$

The angle θ represents argument or phase of the complex number z. Modulus of the complex number represents the amplitude of the oscillation.

  - $x(t)$ and $y(t)$ projections are the real and imaginary parts of $z(t)$:

$$x(t) = \mathrm{Re}[Ae^{i\omega t}], \qquad y(t) = \mathrm{Im}[Ae^{i\omega t}].$$

1.2 Kinematics of SHM

- Velocity and acceleration of SHM. Given the displacement of a SHM given by

$$x(t) = A\cos(\omega t + \alpha),$$

velocity and acceleration ca be easily found:

$$\begin{aligned} &\text{velocity:} && v(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t} &&= -A\omega\sin(\omega t + \alpha) \\ &\text{acceleration:} && a(t) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2} = &&\quad A\omega^2\cos(\omega t + \alpha) = -\omega^2 x. \end{aligned} \tag{1.3}$$

- In the phasor notation, geometry of the velocity and acceleration are easy to visualize. We can do similar calculations for the y-

DRAFT

DRAFT

projection of the phasor:

$$v_y = A\omega\cos(\omega t+\alpha), \qquad a_y = -A\omega^2\sin(\omega t+\alpha) = -\omega^2 y.$$

- **Calculating velocities and accelerations very efficiently**.

  Calculations of both the x- and y-projections can be done accomplished very efficiently in one go if we use the complex phasor representation instead.

  - For some problems, working with complex numbers is easier than working with vectors. Instead of working with multiple components, one at a time, we need to deal with only one complex scalar, that too mostly in the exponential form, which is the easiest mathematical function to deal with. The additional book keeping we need to do is the complex number i in the exponent.
  - Start with

    $$z(t) = x(t) + i\,y(t) = A\left[\cos(\omega t+\alpha) + i\,\sin(\omega t+\alpha)\right] = Ae^{i(\omega t+\alpha)}.$$

    It's the simplest function to take the derivative in the whole of mathematics.
  - Then, applying chain rule, phasor velocity becomes:

    $$\tilde{v} = \frac{\mathrm{d}z}{\mathrm{d}t} = Ae^{i(\omega t+\alpha)}\frac{\mathrm{d}}{\mathrm{d}t}[i(\omega t+\alpha)] = i\omega Ae^{i(\omega t+\alpha)} = i\omega z. \quad (1.4)$$

    This means:

    $$\tilde{v} = \frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(x+iy) = \frac{\mathrm{d}x}{\mathrm{d}t} + i\frac{\mathrm{d}y}{\mathrm{d}t} = i\omega(x+iy) = -\omega y + i\omega x \qquad \Rightarrow$$

    Taking the real and imaginary parts, we immediately obtain the velocities of both x- and y- components of the uniform circular motion in one stroke:

    $$\frac{\mathrm{d}x}{\mathrm{d}t} = -\omega y = -\omega A\sin\omega t; \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \omega x = \omega A\cos\omega t.$$

  - The phasor acceleration is obtained by repeating the process:

    $$\tilde{a} = \frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = \frac{\mathrm{d}\tilde{v}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left[i\omega Ae^{i(\omega t+\alpha)}\right] = A(i\omega)(i\omega)e^{i(\omega t+\alpha)} = i^2\omega^2 Ae^{i(\omega t+\alpha)} = -\omega^2 Ae^{i(\omega t+\alpha)} \Rightarrow \quad (1.5)$$

    $$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = -\omega^2 z. \quad (1.6)$$

    This means:

    $$\frac{\mathrm{d}^2 z}{\mathrm{d}t^2} = \frac{\mathrm{d}^2}{\mathrm{d}t^2}(x+iy) = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + i\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = -\omega^2(x+iy) \qquad \Rightarrow$$

DRAFT

DRAFT

taking the real and imaginary parts, we immediately accelerations of both x- and y- components of the uniform circular motion in one stroke:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -\omega^2 y.$$

- ◇ The only difficult math involved in the calculations above is by multiplication by i for obtaining the projections of velocity of the uniform circular motion.
- ◇ Using Eq.(1.4), phasor acceleration can be related to the phasor velocity by

$$\frac{\mathrm{d}\tilde{v}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}[i\omega z] = i\omega\frac{\mathrm{d}z}{\mathrm{d}t} = i\omega\tilde{v}.$$

Thus, what was two second order ODE now becomes a single first order ODE!

- **Geometrical significance**.
  - ◇ Velocity: the phasor velocity in Eq. (1.4) can be written more conveniently if we make use of the polar form form of of i:

$$e^{i\pi/2} = \cos(\pi/2) + i\,\sin(\pi/2) = i.$$

Then,

$$\tilde{v} = i\omega A e^{i(\omega t+\alpha)} = \omega A e^{i\pi/2} e^{i(\omega t+\alpha)} = \omega A e^{i(\omega t+\alpha+\pi/2)} = \omega A e^{i(\theta+\pi/2)}.$$

We see that the velocity has the amplitude ωA which is ω times that of the displacement, with phase increased by $\pi/2$.

  - ◇ Similarly, for acceleration,

$$\tilde{a} = -\omega^2 z = \omega^2 A e^{i(\omega t+\alpha+\pi)} \qquad \text{or} \qquad \vec{a} = -\omega^2\vec{r}$$

which means that $\vec{a}$ is exactly in the direction opposite to that of the position vector. Or, equivalently, the acceleration phasor has phase π more than that of the position phasor.

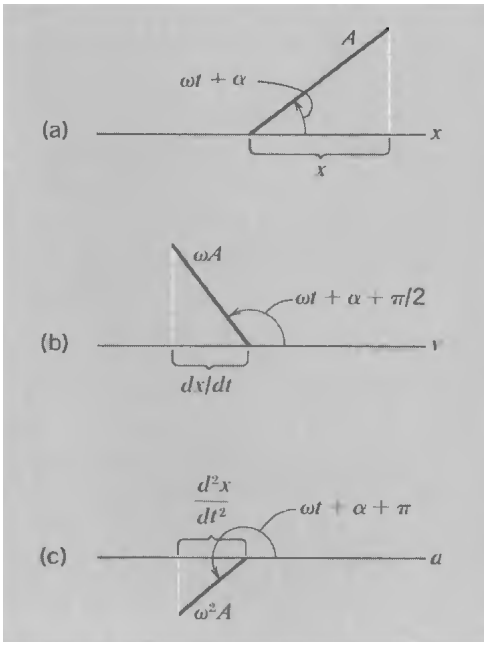


Fig. 1.3: (a) Displacement vector (phasor) z and its real projection x. (b) Velocity vector dz/dt and its real projection dx/dt. (c) Acceleration vector d^2z/dt^2 and its real projection d^2x/dt^2.

- **Exercise 1.1.** A mass on the end of a spring oscillates with an amplitude of 5 cm at a frequency of 1 Hz (cycles per second). At $t = 0$ the mass is at its equilibrium position ($x = 0$).

  (a) Find the possible equations describing the position of the mass

DRAFT

DRAFT

as a function of time, in the form $x = A\cos(\omega t + \alpha)$, giving the numerical values of A, ω, and α.

(b) What are the values of x, dx/dt, d^2x/dt^2 at $t = \frac{8}{3}$ sec?

- **Exercise 1.2.** A point moves in a circle at a constant speed of 50 cm/sec. The period of one complete journey around the circle is 6 sec. At $t = 0$ the line to the point from the center of the circle makes an angle of 30° with the x axis.

(a) Obtain the equation of the x coordinate of the point as a function of time, in the form $x = A\cos(\omega t + \alpha)$, giving the numerical values of A, ω, and α.

(b) Find the values of x, dx/dt, d^2x/dt^2 at $t = 2$ sec.

1.3 The Spring-Mass System, its ODE and Solutions

- Let us recall from Eq.(1.3) that a SHM given by $x(t) = A\cos(\omega t + \alpha)$ or $A\sin(\omega t + \phi)$ obeys the linear 2nd order ODE

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x. \tag{1.7}$$

- What is the simplest physical system they obeys this 2nd order linear ordinary differential equation (ODE)? All of us know that the familiar spring-mass system as shown in the Fig.1.4 obeys the same ODE.

- When the mass connected to a spring fixed to the wall on the left is displaced from its equilibrium position $x = 0$ by distance x, the restoring force on the mass from the spring is

$$F = -kx,$$

where k is known as the spring constant. This is known as the Hooke's law.

- By Newton's 2nd law, equation of motion of the attached mass is

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F = -kx. \tag{1.8}$$

- Dividing Eq.(1.8) by mass we get

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{k}{m}x.$$

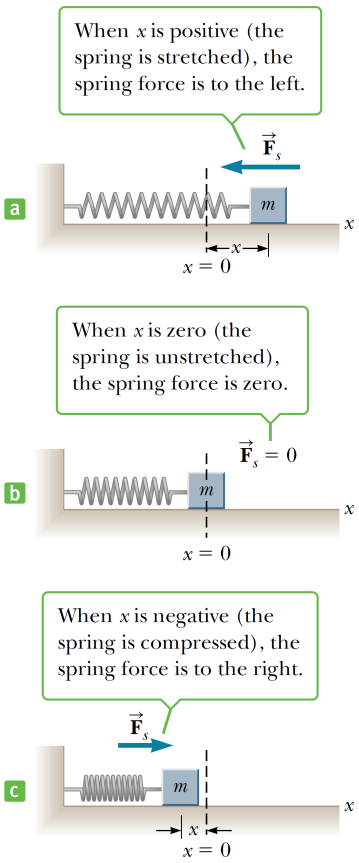


Fig. 1.4: The force exerted by a spring on an object varies with the displacement of the object from the equilibrium position, $x = 0$.

DRAFT

DRAFT

By comparing with Eq.(1.7), we obtain the relationship between, k, m and ω, and the angular frequency, period, etc:

$$\boxed{\omega^2 = \frac{k}{m},} \qquad \boxed{\omega = \sqrt{\frac{k}{m}},} \qquad \boxed{T = 2\pi\sqrt{\frac{m}{k}}.}$$

- Notice that

  $$\omega^2 = \frac{k}{m} = \frac{|F/x|}{m} = \text{restoring force per unit displacement per unit mass.}$$

  Many oscillating system that will encounter can be brought to this form and we will be able to calculate the effective spring constant and the angular frequency.

1.3.1 Solution to the ODE of Spring-Mass System – Method 1

- We already know that $\sin\omega t$ and $\cos\omega t$ satisfy the SHM ODE Eq. (1.7)

  $$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\omega^2 x. \tag{1.9}$$

  However we will approach the SHM functions from the point of view of solving a 2nd order linear ODE.

- It is important to recognize that it is because the force is **linear** in x, we have a linear ODE which is easily solvable.

- **The simplest method of solution** is to assume that the solution, or the trial solution, is of the form

  $$x(t) = Ce^{\lambda t}. \tag{1.10}$$

  - ◇ Why? This is because, $e^{\lambda t}$ is the only function that retains its form under the action of one or more differentiation operations.
  - ◇ The task is to find the parameters λ and C. Substituting (1.10) into (1.9) gives us what is known as the characteristic equation

    $$(\lambda^2 + \omega^2)Ce^{\lambda t} = 0 \quad \text{for all } t \Rightarrow \qquad \lambda^2 + \omega^2 = 0 \quad \Rightarrow$$

    $$\lambda_{1,2} = \pm i\omega.$$

  - ◇ The two independent solutions are

    $$x_1(t) = e^{i\omega t}, \qquad x_2(t) = e^{-i\omega t}.$$

    The general solution is a linear combination of the two indepen-

DRAFT

DRAFT

dent solutions

$$\boxed{x_1 = c_1 e^{i\omega t} + c_2 e^{-i\omega t}.} \tag{1.11}$$

The coefficients$c_{1,2}$ are the constants to be determined using initial conditions.

- **Solution as linear combination of sinusoidal functions**.
  - Using the Euler formula linking exponential function and trigonometric functions,

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{1.12}$$

the general solution (1.11) can also be expressed in the form

$$\boxed{x(t) = A_1 \cos\omega t + B_1 \sin\omega t,} \tag{1.13}$$

where the new undetermined constants A_1, B_1 are

$$A_1 = c_1 + c_2, \qquad B_1 = i(c_1 - c_2), \tag{1.14}$$

In this chapter, we use the subscripted symbols to avoid confusion with the amplitude A.

  - Given that x, $\sin\omega t$ and $\cos\omega t$ are real numbers for all t, so are A_1 and B_1.

- **Solving the coefficients A_1 and B_1 given the initial conditions**:
  - Let the initial conditions at time $t = 0$ be

$$x(0) = x_0, \qquad v(0) = \dot{x}(0) = v_0. \tag{1.15}$$

Applying these conditions, we arrive at

$$\boxed{A_1 = x_0, \qquad B_1 = v_0/\omega.} \tag{1.16}$$

  - Thus, the solution takes the form

$$\boxed{x(t) = x_0 \cos\omega t + \frac{v_0}{\omega}\sin\omega t.} \tag{1.17}$$

  - It can be expressed in the form standard amplitude-phase form:

$$\boxed{x(t) = A\cos(\omega t + \alpha).} \tag{1.18}$$

  - Expanding Eq.(1.18) and comparing with Eq.(1.17), we obtain

$$A\cos\alpha = x_0, \qquad A\sin\alpha = -v_0/\omega,$$

DRAFT

DRAFT

from which we arrive at

$$\boxed{A = \left(x_0^2 + v_0^2/\omega^2\right)^{1/2}, \qquad \alpha = \tan^{-1}\left(\frac{-v_0}{\omega x_0}\right).}$$

To get the correct value of α in $[0, 2\pi)$ you must find the quadrant of the point

$$\boxed{(A\cos\alpha,\, A\sin\alpha) = (x_0, -v_0/\omega).}$$

- **Exercise 1.3.** Write the solution in the form

$$x(t) = A\sin(\omega t + \phi)$$

and show that

$$\phi = \tan^{-1}\left(\frac{\omega x_0}{v_0}\right).$$

To get the correct value of ϕ in $[0, 2\pi)$ one must find the quadrant of the point

$$(A\cos\phi,\, A\sin\phi) = (v_0/\omega, x_0).$$

- **General solution form-2**.
  - ◇ From (1.14), solving for $c_{1,2}$ we can express $c_{1,2}$ in terms of A_1, B_1 :

$$c_1 = \frac{1}{2}(A_1 - iB_1), \qquad c_2 = \frac{1}{2}(A_1 + iB_1).$$

  - ◇ Since A_1, B_1 are real, we must have

$$c_2 = c_1^* \qquad \text{(complex conjugate of } c_1\text{)}.$$

  - ◇ Thus, we can write (1.11) in the form

$$\boxed{x = c_1 e^{i\omega t} + c_1^* e^{-i\omega t}.} \tag{1.19}$$

  - ◇ It says that *the general solution can be expressed as a sum of a phasor $e^{i\omega t}$ rotating in the counter-clockwise sense and its complex conjugate phasor $e^{-i\omega t}$ that rotates in the clockwise wise*. They represent two independent solutions of the ODE.
  - ◇ From $x = A\cos\theta$ and $y = A\sin\theta$, we can express the complex

DRAFT

DRAFT

number $z = x + iy$ in the important polar form

$$z = A(\cos\theta + i\sin\theta) = Ae^{i\theta}.$$

Similarly, we can write

$$c_1 = |c_1|e^{i\alpha} = Ce^{i\alpha},$$

where $C = |c_1|$ is a positive real number, and α, also a real number, is the phase or argument of the complex number. The the solution (1.19) can now be expressed in the form

$$\begin{aligned} x &= Ce^{i(\omega t+\alpha)} + Ce^{-i(\omega t+\alpha)} && (1.20)\\ &= 2C\cos(\omega t+\alpha). && (1.21)\end{aligned}$$

◇ Finally, defining $A = 2C$, we obtain the standard form:

$$\boxed{x = A\cos(\omega t+\alpha).} \qquad (1.22)$$

◇ What is the connection between the phase ϕ in (1.18) and the phase α in (1.22)? It's quite simple:

$$\alpha = \phi - \pi/2, \qquad \text{or} \qquad \phi = \alpha + \pi/2.$$

- **Example 1.1.** Start a SHM with $\omega = 1$ with initial conditions $x_0 = 1$, and $v_0 = -1$ (moving towards the equilibrium point from the positive side). An illustration has been shown in Fig.(1.5). Find A, α.

  ◇ Simple calculations give us

$$A = \sqrt{2}, \qquad \alpha = \tan^{-1}\left(\frac{-(-1)}{1\times 1}\right) = \pi/4.$$

  ◇ But, how do we know α is not $5\pi/4$, since $\tan(5\pi/4) = 1$ also? In the 3rd quadrant of the circular motion, the x-velocity $\dot{x}$ of the oscillating mass is positive, where as in our case, $\dot{x}(0) = -1 < 0$ (is negative) as shown in Fig.1.5.

  ◇ Hence the solution is

$$x(t) = \sqrt{2}\cos(t + \pi/4).$$

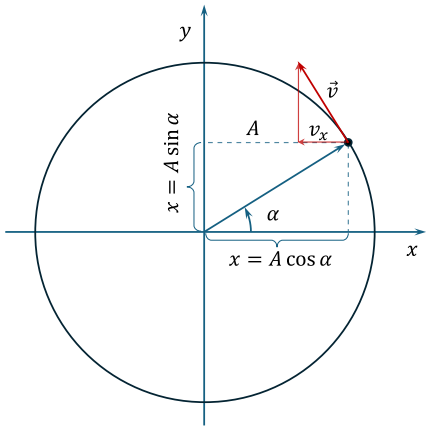


Fig. 1.5: Initial state of an SHM.

DRAFT

DRAFT

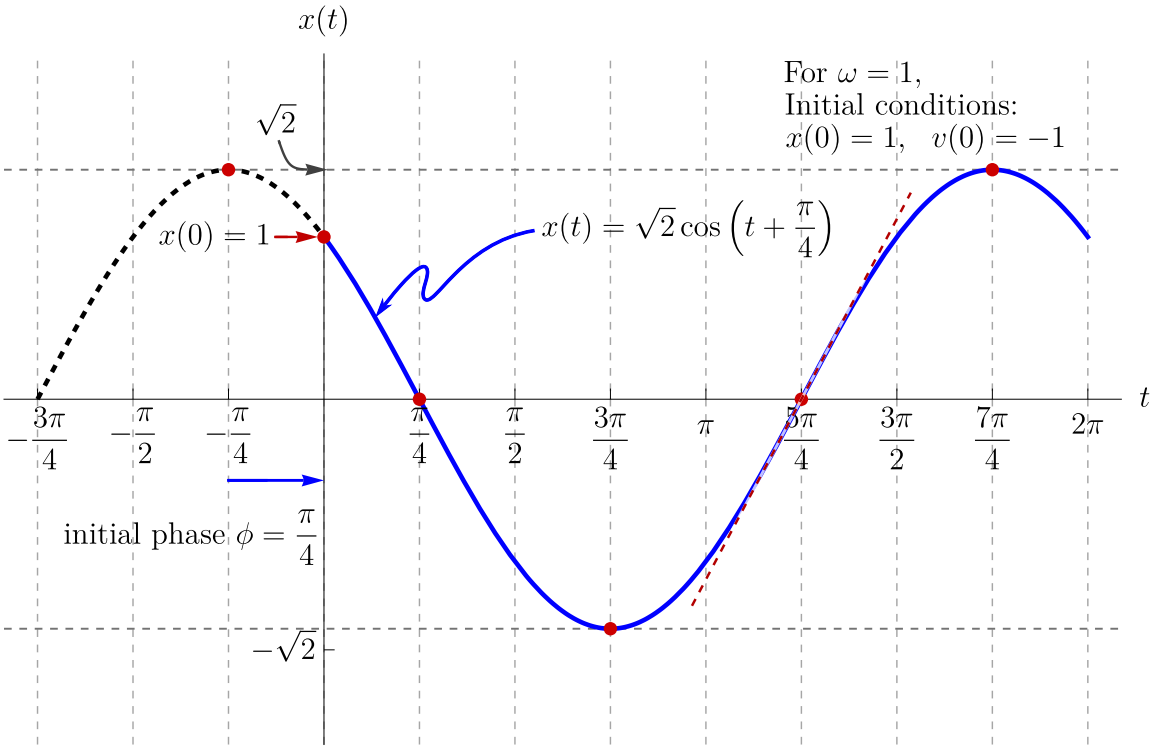


Fig. 1.6: A plot of $x(t) = \sqrt{2}\cos(t + \pi/4)$.

- **Exercise 1.4.** (a) For frequency $\nu = 1$, and initial conditions $x(0) = x_0 = 1$, $v(0) = v_0 = 2\pi$, show that the solution is

$$x(t) = \sqrt{2}\sin(2\pi t + \pi/4).$$

Hint: $\cos(\theta - \pi/2) = \cos(\pi/2 - \theta) = \sin\theta$. (b) Plot the solution. (c) What are (i) amplitude, (ii) angular frequency, (iii) period?

1.4 Kinetic, Potential and Total Energies and the Energy Method

1.4.1 Kinetic, potential and total energies, average kinetic and potential energies

See Berkeley Physics Vol.1, Mechanics, 2nd Edition, Chapter 7, pages 213–214

1.4.2 The energy method of solution to equation of motion.

See Berkeley Physics Vol.1, Mechanics, 2nd Edition, Chapter 7, pages 204, 206

DRAFT

DRAFT

1.5 Simple harmonic motions and natural frequencies of simple physical systems

1.5.1 Simple pendulum

See the introductory section in uploaded notes on coupled oscillations.

1.5.2 Elastic bar

- Elasticity of metals is the basis of elastic nature of common springs you encounter in every day life. It is important to appreciate this property of metals.
- Stress σ and strain ϵ in an elastic bar

$$\sigma = \frac{F_{\text{ext}}}{A}, \qquad \epsilon = \frac{\Delta L}{L}.$$

- Hooke's law for an elastic bar: strain induced is proportional to the applied stress, or stress developed in the bar is proportional to strain imposed:

$$\boxed{\sigma = Y\epsilon.}$$

  Here Y is a constant of the elastic solid called the *Young's modulus*.
- Putting in the definitions of σ and ϵ into the Hooke's law for the elastic bar, we get

$$F_{\text{ext}} = \frac{AY}{L}\Delta L.$$

- Let $x = \Delta L$ the displacement associated with the extension or compression of the bar. The restoring force from bar on an externally attached mass is simply negative of the above expression and can be expressed in the standard form

$$F = -F_{\text{ext}} = -\left(\frac{AY}{L}\right)x = -kx.$$

- Thus the natural frequency of a mass m attached to an elastic bar is

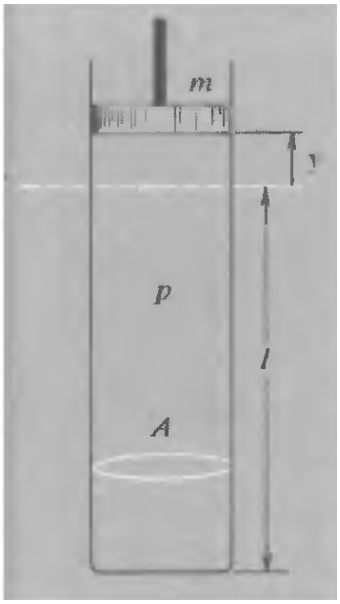
$$\omega = \sqrt{\frac{AY}{mL}}.$$

DRAFT

DRAFT

1.5.3 Oscillations of piston in a cylinder filled with air

- References
  - ◇ A.P. French, Vibrations and Waves, Ch.3, pages 57–59.
  - ◇ Lectures



Key Ideas:

- Piston floats in a cylinder filled with air balanced by the pressure in the cylinder in excess of external atmospheric pressure. Pressure inside the cylinder is

$$P = P_{\text{atm}} + \frac{mg}{A},$$

  where m is the mass of the piston, A is the cross-sectional area of the cylinder, as illustrated in the figure on the right.

- **Mechanism of restoring force for piston oscillations**. When the piston is displaced from equilibrium, say by moving upward by distance y, there is a drop in the pressure inside the cylinder, and, consequently, external pressure pushes the piston down. This is mechanism of the restoring force in this system. When the piston is pushed down, pressure inside the cylinder becomes larger, and it tends to push the piston upwards to restore the equilibrium.

- Displacement of piston by a y from the equilibrium position causes a change in volume of the air column by

$$\Delta V = Ay.$$

- **Change in pressure of the air when the piston is displaced** inside the piston depends on the relationship between the pressure P and volume V inside the cylinder.
  - ◇ Isothermal changes: if the piston moves so slowly that air is in thermal equilibrium with the surrounding air at fixed temperature T_{atm}, then, P and V are related by

$$PV = nRT = \text{a constant}.$$

    Here, n and T denote the number of moles and the temperature of the air in the cylinder, and $R = 8.314$ Joules/Kelvin.

  - ◇ Adiabatic changes. Typically, air mostly has not much time to exchange heat with the surroundings and the changes are adiabatic. So, the relationship between them must be adiabatic pro-

DRAFT

DRAFT

cess relation:

$$PV^{\gamma} = C = \text{a constant},$$

where $\gamma = C_P/C_V$ is the ratio of specific heats of the gas inside the cylinder.

- Thus, in general, we may write

$$PV^{r} = C, \tag{1.23}$$

where

$$r = \begin{cases} 1, & \text{for isothermal changes} \\ \gamma & \text{for adiabatic changes} \end{cases}$$

- **Calculating the pressure changes when volume of the piston changes**. Let $P_0$and V_0 be equilibrium pressure and volume connected by

$$P_0 V_0^{r} = C.$$

At any position of the piston let

$$\Delta P = P - P_0, \quad \Delta V = V - V_0.$$

Taking the differential of the relation (1.23), we get

$$\Delta P V^{r} + P\Delta(V^{r}) = 0 \quad \Rightarrow \qquad \Delta P + rPV^{r-1}\Delta V = 0 \quad \Rightarrow$$

$$\Delta P = -r\frac{PV^{r-1}}{V^{r}}\Delta V = -\frac{rP}{V}\Delta V = -\frac{rP}{V}Ay.$$

Actual pressure and volume are very close to the equilibrium values, and all changes are accounted for already with the appearance of y in the equation above. So we may replace $P = P_0$ and $V = V_0$ and write

$$\boxed{\Delta P \approx -\frac{rP_0}{V_0}Ay.}$$

- Calculating the restoring force on the piston.
  - In the final step recognize that the restoring force is indeed the pressure change times the area:

$$\boxed{F = A\Delta P = -\frac{rP_0A^2}{V_0}y = -ky,}$$

DRAFT

DRAFT

with effective spring constant given by

$$\boxed{k = \frac{rP_0A^2}{V_0}.}$$

- Frequency of oscillations, therefore, is

$$\boxed{\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{rP_0A^2}{mV_0}}.}$$

- For adiabatic oscillations,

$$\boxed{\omega = \sqrt{\frac{\gamma P_0A^2}{mV_0}}.}$$

- **Exercise 1.5.** Assume $P_0 = 1\,\text{atm} \approx 1.01 \times 10^5\,\text{Pa}$, $V_0 = 1\,\text{litre} = 10^{-3}\,\text{m}^3$, $A = 20\,\text{cm}^2 = 2 \times 10^{-3}\,\text{m}^3$, $m = 0.25\,\text{kg}$, $\gamma = 7/5$. Find the frequency $\nu = \omega/(2\pi)$ and the period $T = 1/\nu$. Assume adiabatic changes. By what factor much will your answer differ if you assumed isothermal changes?

1.5.4 LC circuit

- Berkeley Physics Vol.1, Mechanics, 2nd Edition, Chapter 7, pages 209–212
- Lectures

1.6 Harmonic approximation of general potentials near minima

- **References**
  - ◇ Berkeley Physics Vol.1, Mechanics, 2nd Edition, Chapter 7, pages 212–213
  - ◇ Berkeley Physics Vol.1, Mechanics, 2nd Edition, Chapter 9, pages 293–294.
- **Key ideas:**
  - ◇ For any arbitrary potential energy curve $U(r)$ that has a minimum at r_0, we can Taylor expand the potential abut the mini-

DRAFT

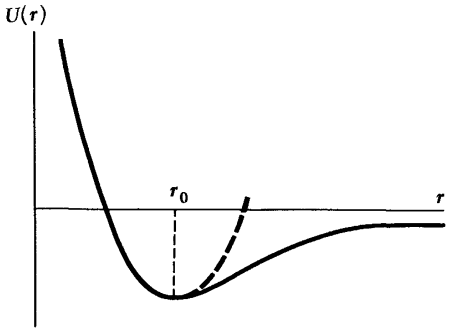
DRAFT

mum point as

$$U(r) = U(r_0) + \frac{1}{2}(r - r_0)^2 U''(r_0) + \cdots,$$

where r_0 is the relative distance between the atoms of the molecules at which the potential is minimum; i.e.,

$$\left(\frac{\mathrm{d}U}{\mathrm{d}r}\right)_{r=r_0} = U'(r_0) = 0,$$



Graph of the potential energy as a function of distance between two atoms combined in a molecule. The position of equilibrium is at r_0. The dashed curve shows the parabola corresponding to a quadratic potential energy function given in Eq. (1.24).

- An example is the potential energy between bound atoms in a molecule as shown in the figure on the right.
- *For small oscillations*, potential energy associated with the relative distance r between molecules near its minimum can be approximated in the form

$$U(r) = \frac{1}{2}K(r - r_0)^2, \tag{1.24}$$

and K is the effective spring constant given by

$$K = \left(\frac{\mathrm{d}^2U}{\mathrm{d}r^2}\right)_{r=r_0} = U''(r_0).$$

- Effective mass in the relative motion is reduced mass of the two atoms in relative motion:

$$\frac{1}{\mu} = \frac{1}{m_A} + \frac{1}{m_B}.$$

- Equation of motion is

$$\mu\frac{\mathrm{d}^2r}{\mathrm{d}t^2} = -\frac{\mathrm{d}U}{\mathrm{d}r} = -K(r - r_0).$$

- By letting $r - r_0 = x$, we arrive at the standard form

$$\mu\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -Kx.$$

- Frequency of oscillations:

$$\boxed{\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{U''(r_0)}{\mu}}.}$$

- **Exercise 1.6.** Work out the example given in Berkeley Physics Vol.1, Mechanics, 2nd Edition, Chapter 9, pages 293–294, and find the frequency of oscillations of a HCl molecule.

DRAFT

DRAFT

1.7 Damped Oscillations, Quality Factor

- **References**
  - Berkeley Physics Vol.1, Mechanics, 2nd Edition, Chapter 7, pages 217-221
  - A.P. French, Vibrations and Waves, Chapter 3, pages 62–70.
  - Lectures

1.8 Driven Oscillations and Resonance

- **References**
  - A.P. French, Vibrations and Waves, Chapter 4, pages 77–92, 96–104, 108–110.
  - Lectures

1.9 Linearity and Superposition Principle

- Linear homogeneous differential equations have the following very interesting and important property:
  - *The sum of any two solutions of linear differential equation is itself a solution.*
- Nonlinear equations do not have do not have thus property*:*
  - *The sum of the two solutions of a nonlinear differential equation is not itself is a solution.*

Exercise 1.7. Consider a linear ODE:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -Cx. \tag{1.25}$$

(a) Let $x_1(t)$ be a solution of the ODE (1.26) for some initial conditions $x_1(0) = x_{10}$ and $\dot{x}_1(0) = \dot{x}_{10}$, and $x_2(t)$ be another solution for some other initial conditions $x_2(0) = x_{20}$ and $\dot{x}_2(0) = \dot{x}_{20}$.

Show that the sum these solutions,

$$x(t) = x_1(t) + x_2(t),$$

with respective initial conditions summed

$$x(0) = x_{10} + x_{20} \qquad \text{and} \qquad \dot{x}(0) = \dot{x}_{10} + \dot{x}_{20},$$

DRAFT

DRAFT

is also solution to the ODE Eq.(1.25)

(b) Show that the scalar multiple of $x_1(t)$, given by

$$x(t) = cx_1(t),$$

with initial conditions

$$x(0) = cx_{10} \qquad \text{and} \qquad \dot{x}(0) = c\dot{x}_{10},$$

is also a solution to the Eq.(1.25).

Exercise 1.8. Consider now a non-linear ODE:

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -Cx + C_2x^2. \tag{1.26}$$

As in the exercise (1.7), again let $x_1(t)$ and $x_2(t)$ be solutions of the ODE (1.26) with the same initial conditions as mentioned therein. Show that their sum $x(t) = x_1(t) + x_2(t)$ is not a solution to the non-linear ODE Eq.(1.26).

Exercise 1.9. Consider a linear inhomogeneous ODE:

$$\ddot{x} + \gamma\dot{x} + Cx = F(t). \tag{1.27}$$

Here, $F(t)$ represents external driving forces.

1. Let x_1 and x_2 be the particular solutions for some external forces $x_1(t)$ and $x_2(t)$ respectively. Show that the linear superposition $x_1(t) + x_2(t)$ is a solution to the ODE with the driving force that is the same linear superposition of these forces $x(t) = x_1(t) + x_2(t)$.

2. As a corollary, apply this result to the case in which $F_2 = 0$ corresponding the homogeneous linear ODE. Its solution may be denoted by $x_2(t) = x_t(t)$, referred to as the transient solution which decays to zero when damping is present.

DRAFT